MILITARY SPECIFICATION

TEST SET, VIBRATION
TTK-402/E24T-148
AND
CABLE CASE
CNK-357/E24T-148

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. The vibration test set TTK-402/E24T-148 and the cable case CNK-357/E24T-148 are used in conjunction with peculiar adapters to monitor engine vibration of a TF34 engine installed in an S-3A airframe. The test set measures mechanical vibration in displacement or velocity units.

1.1.1 Major components list. The major components are:

Component	Part Number	Quantity
TTK-402/E24T-148	1361AS200-1	1
CNK-357/E24T-148	1361AS700-1	1

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD) Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:11Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

Military

MIL-Q-9858	Quality Program Requirements
MIL-E-17555	Electronic and Electrical Equipment, Accessories, and Repair Parts: Packaging and Packing of
MIL-T-18303	Test Procedures, Reproduction, Acceptance, and Life for Aircraft Electronic Equipment, Format for
MIL-T-28800	Test Equipment for Use with Electrical and Electronic Equipment, General Specification for

STANDARDS

Military

MIL-STD-108	Definition of and Basic Requirements for Enclosures for Electric and Electronic Equipment
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-471	Maintainability Demonstration
MIL-STD-781	Reliability Tests: Exponential Distribution
MIL-STD-810	Environmental Test Methods

Source: https://assist.dla.mil Downloaded: 2026-10-23T11:11Z
Check the source to verify that this is the current version before use.

HANDBOOKS

Military

(C) MIL-HDBK-235-2 Electromagnetic (Radiated) Environment Considerations for Design and Procurement of Electrical and Electronic Equipment

Handbook may be obtained from:

Commander
Naval Electronic Systems Command
Standardization Program Branch
Washington, DC 20360

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

Naval Air Systems Command

DL 1361AS100 Test Set, Vibration A/E24T-148

Drawings may be obtained from:

Commanding Officer
Naval Air Engineering Center
Attention: Code 92611
Lakehurst, NJ 08733

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.4 and 6.2).

3.2 Design and construction. The test set shall be designed in accordance with Naval Air Systems Command drawing number DL 1361AS100.

3.2.1 Controls and switches. Controls and switches shall be located in accordance with Figure 1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:11Z
Check the source to verify that this is the current version before use.

3.2.2 Test set classification. The vibration test set and cable case shall meet the requirements of MIL-T-28800 for Type I, Class 2, Style A equipment, except as otherwise specified herein.

3.2.3 Input power. The test set shall be capable of operating from a 115 ±10% VRMS, 50-420 Hz source.

3.2.4 Readout. The readout display on the test set shall indicate velocity or displacement as selected by the "MODE SELECT" rotary switch. A visual annunciator shall be provided to indicate the mode of operation.

3.2.5 Open/short circuit indication. The display shall be blank when resistance across input terminals is greater than 5000 ohms or less than 100 ohms. The display shall not be blank for resistance values of 200 to 2000 ohms.

3.2.6 Front panel lighting/display lighting (PNL LT/DSPL) switch. The front panel lighting/display lighting switch controls the brightness of the illuminated front panel and the display. In the "OFF-BRT" position, the front panel is not illuminated and the display is bright. In the "ON-DIM" position, the front panel is illuminated and the display is dim.

3.2.7 "FILTER SELECT" switch. The "FILTER SELECT" rotary switch controls the selection of any one of four filter positions or an unfiltered signal.

3.2.8 Display test (DSPL-TEST) switch. The "DSPL-TEST" switch activates all segments of the digital display and the visual annunciator.

3.2.9 Function test (FCTN-TEST) switch. The "FCTN-TEST" switch causes an internally generated simulated vibration signal to be displayed on the vibration readout. The display shall indicate 55X (where X can be any number) for the following conditions:

"INPUT SELECT" switch on "1"
"FILTER SELECT" switch on "OUT"
"MODE SELECT" switch on "D" or "V"
"SENSITIVITY SELECT" switch on "60," "105," "110," "122," "135" or "145"

3.2.10 "MODE SELECT" switch. The "MODE SELECT" switch is used to select either velocity or displacement mode of operation.

3.2.11 "SENSITIVITY SELECT" switch. The "SENSITIVITY SELECT" switch allows the operator to match the gain to any of six velocity transducer sensitivities.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:11Z
Check the source to verify that this is the current version before use.

3.2.12 Power (PWR) switch and indicator. When 115 VAC 50-420 Hz power is connected to the test set, the "PWR" switch shall turn the test set "on" and "off" and the indicator shall glow red when power is on.

3.2.13 "ELAPSED TIME (HRS)" indicator. The "ELAPSED TIME (HRS)" indicator shall be in accordance with Requirement 51 of MIL-STD-454.

3.2.14 "INPUT SELECT" switch. The "INPUT SELECT" rotary switch allows the operator to select any one of six input signals from six velocity transducers having nominal coil resistances of 1150 ohms or less and nominal sensitivities of 60 to 145 millivolt/inch/second.

3.2.15 Interchangeability. The vibration test set shall meet the requirements of MIL-T-28800.

3.3 Performance.

3.3.1 Displacement. The readout display shall indicate the amplitude of the displacement within the limits shown below.

Range	0-19.9 mils peak-to-peak
Resolution	0.1 mils peak-to-peak
Accuracy	±3% of full scale (0.6 mils)

3.3.2 Velocity. The readout display shall indicate the amplitude of the velocity within the limits shown below.

Range	0-59.9 in./sec average
Resolution	0.1 in./sec
Accuracy	±3% of full scale (1.8 in./sec)

3.3.3 Linearity. The linearity of the test set shall be within ±1% of full scale.

3.3.4 Noise. The noise level of the amplifier shall not exceed ±1% of full scale when the input is terminated in a resistive load of 470 ±5% ohms.

3.4 Environmental. The vibration test set and cable case shall meet the performance requirements of 3.3 before and after testing when exposed to the following environmental conditions and tested in accordance with the applicable test method.

3.4.1 Temperature and altitude.

a. Operating temperature range shall be -30°F to +130°F.

b. Operating altitude shall be 10,000 foot maximum.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:11Z
Check the source to verify that this is the current version before use.

c. Non-operating temperature range shall be -65°F to +160°F.

d. Non-operating altitude shall be 50,000 foot maximum.

3.4.2 Vibration. The test set shall meet the vibration requirements of MIL-T-28800 for Class 6 equipment.

3.4.3 Mechanical shock. When tested in accordance with 4.6.5.3, the test set shall meet the performance requirements of 3.3.

3.4.4 Water resistance (splashproof). Water resistance shall be in accordance with the requirements of MIL-T-28800 for Style A.

3.4.5 Humidity. The test set shall be tested in accordance with 4.6.5.5.

3.4.6 Salt atmosphere. The test set shall meet the exposure, enclosure requirements of MIL-T-28800 for Style A.

3.4.7 Explosive atmosphere. The test set shall meet the requirements of MIL-T-28800.

3.4.8 Sand and dust. The test set shall meet the requirements of MIL-T-28800.

3.4.9 Fungus resistance. The vibration test set shall meet the fungus resistance requirements of MIL-T-28800. In addition, a certification of non-nutrient fungus-inert materials duly witnessed and certified shall be submitted to the contracting activity for approval.

3.4.10 Electromagnetic interference. The test set shall meet the requirements of MIL-T-28800 for Class 2 equipment, with the additional requirements of Conducted Emissions CE01, CE02,CE04, and Radiated Susceptibility RS01, RS02, and RS03 at 200 volts/meter.

3.4.11 Electromagnetic vulnerability. The vibration test set shall not be vulnerable to the electromagnetic environment of an aircraft carrier flight deck when tested in accordance with 4.6.5.11.

3.4.12 Reliability. The vibration test set shall have a Mean-Time-Between Failure (MTBF) of 295 hours demonstrated at the 95% reliability level when tested in accordance with 4.6.6.

3.4.13 Mean Time to Repair/inherent availability (MTTR/Ai). The vibration test set shall have an MTTR not to exceed four hours when demonstrated in accordance with 4.6.17. The minimum acceptable availability is 97%. Inherent (design) availability goal is 99%.

3.6 Workmanship. Workmanship shall be in accordance with Requirement 9 of MIL-STD-454.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:11Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in the applicable test method document or applicable paragraph(s) of this specification.

4.4 First article.

4.4.1 First article inspection. The first article inspection shall consist of examinations and tests specified in table I.

4.4.2 Sample size. The contract or purchase order shall define the sample size. The samples shall be manufactured using production methods that will be employed for the balance of the contract (see 6.2).

4.4.3 Failure criteria. Failure to meet the requirements of any tests or examinations of table I shall constitute failure of the first article inspection unless the contracting activity, upon review of the test plans and reports, provides the supplier relief, in writing, for the discrepancies.

4.5 Quality conformance inspection. Tests and examinations for quality conformance inspection are listed below.

a. Preoperational inspection (see 4.5.3).

b. Performance inspection (see 4.5.4).

4.5.1 Sample size. The contract or purchase order shall define the sample size. The samples shall be manufactured using production methods that will be employed for the balance of the contract (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:11Z
Check the source to verify that this is the current version before use.

4.5.2 Test plans and test reports. Unless otherwise specified in the contract or purchase order, test plans and test reports shall be submitted to the contracting activity for approval.

4.5.2.1 Test plans. Test plans shall detail the test procedures, define the required test equipment, document programs for automatic test equipment, and define the criteria for acceptance. Test procedures for maintenance (troubleshooting) may be included.

4.5.2.2 Test reports. Test reports shall summarize the results of the preoperational and performance inspections including the corrective actions required to obtain the acceptance criteria. When the acceptance criteria is waived, justification for the waiver must be included in the test report.

4.5.3 Preoperational inspection. The preoperational inspection of MIL-T-28800 shall be performed.

4.5.4 Performance inspection. The requirements of Section 3.3 shall be demonstrated utilizing the test plans of 4.5.2.1.

4.6 Inspection methods.

4.6.1 Displacement. With a transducer connected to the selected input channel, the displacement indicated shall be within the limits specified in 3.3.1.

4.6.2 Velocity. With a transducer connected to the selected input channel, the velocity indicated shall be within the limits specified in 3.3.2.

4.6.3 Linearity. With the "MODE SELECT" switch in the "D" position, the linearity shall be as indicated in 3.3.3.

4.6.4 Noise. When the input is terminated in a resistive load of 470 ±5% ohms with the "MODE SELECT" switch in the "D" position, the noise level shall not exceed the amount specified in 3.3.4.

4.6.5 Environmental.

4.6.5.1 Temperature and altitude. The temperature and altitude test shall be in accordance with MIL-STD-810, Method 504.1, Equipment Category 2.

4.6.5.2 Vibration. Vibration shall be tested in accordance with MIL-T-28800.

4.6.5.3 Mechanical shock. The test set shall meet the requirements of 3.4.3 when tested in accordance with the pulse shock test of MIL-T-28800.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:11Z
Check the source to verify that this is the current version before use.

4.6.5.4 Water resistance (splashproof). Water resistance shall be tested in accordance with the splashproof, Style A, test of MIL-T-28800.

4.6.5.5 Humidity. The humidity test shall be in accordance with Method 507.1, Procedure I of MIL-STD-810.

4.6.5.6 Salt atmosphere. The test set shall be tested in accordance with the salt atmosphere, exposure, enclosure, test of MIL-T-28800.

4.6.5.7 Explosive atmosphere. The test set shall be subjected to the explosive atmosphere test of MIL-T-28800.

4.6.5.8 Sand and dust. The test set shall be tested in accordance with the sand and dust test, Class 1 and 2, Style A and T, of MIL-T-28800.

4.6.5.9 Fungus resistance. The vibration test set shall meet the requirements of 3.4.9 when tested in accordance with the fungus resistance test of MIL-T-28800.

4.6.5.10 Electromagnetic interference. The test set shall be subjected to the electromagnetic interference test of MIL-T-28800.

4.6.5.11 Electromagnetic vulnerability. The vibration test set shall be interfaced to an S-3A aircraft via its cables and shall perform within the limits specified in 3.3.1 when subjected to the electromagnetic environments of MIL-HDBK-235-2 (Navy) Table Va. Dynamic (engine operating) tests shall be performed by personnel from the Naval Surface Weapons Center, Dalgren, VA and the Naval Air Test Center, Patuxent River, MD.

4.6.6 Reliability. Reliability shall be tested in accordance with MIL-STD-781, Test Plan XXV, Test Level A. Test time shall be extended as required.

4.6.7 Maintainability. Maintainability shall be demonstrated in accordance with MIL-STD-471, Test Method 4.

5. PACKAGING

5.1 Preservation, packaging, packing and marking. Packaging shall be in accordance with levels of MIL-E-17555 specified by the contract or purchase order.

5.2 Delivery of test samples. The disposition of test samples shall be specified in the contract or purchase order.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:11Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use. The vibration test set TTK-402/E24T-148 and cable case CNK-357/E24T-148 are used by organizational (flight line) personnel to monitor the mechanical engine vibrations of the TF34 engine used in the S-3A aircraft. The equipment will be used ashore and afloat, in all environmental conditions.

6.2 Ordering data. Acquisition documents shall specify the following:

a. Title, number, and date of this specification.

b. The levels of preservation, packaging, packing, and marking per MIL-E-17555 (see 5.1).

c. The responsibility for the performance of inspections not to be performed by the supplier (see 4.1).

d. The quantity of samples for first article inspection and quality conformance inspection and the disposition of the samples (see 4.4.2 and 4.5.1).

e. The contractor's test facilities shall be available to the Government inspector(s) at all times during all testing required by this specification.

f. Quality Assurance Program requirement in accordance with MIL-Q-9858 (tailored).

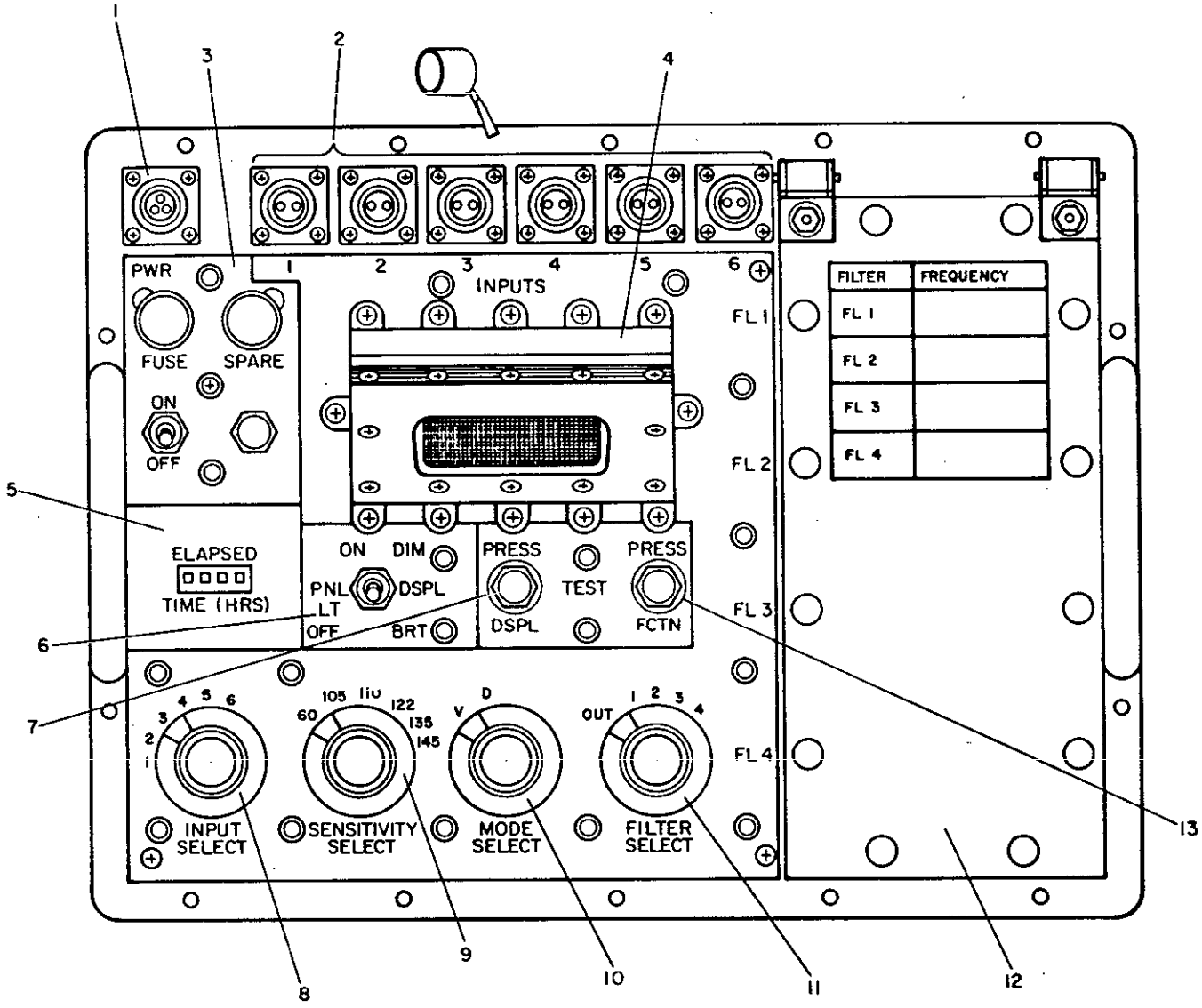
Preparing activity:
Navy - AS
(Project No. 4920-N383)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:11Z
Check the source to verify that this is the current version before use.

TABLE I. First article examinations and tests (see 4.4.1).

Inspection	Paragraph	
	Requirement	Method
Preoperational		4.5.3
Displacement	3.3.1	4.6.1
Velocity	3.3.2	4.6.2
Linearity	3.3.3	4.6.3
Noise	3.3.4	4.6.4
Temperature and altitude	3.4.1	4.6.5.1
Vibration	3.4.2	4.6.5.2
Mechanical shock	3.4.3	4.6.5.3
Water resistance	3.4.4	4.6.5.4
Humidity	3.4.5	4.6.5.5
Salt atmosphere	3.4.6	4.6.5.6
Explosive atmosphere	3.4.7	4.6.5.7
Sand and dust	3.4.8	4.6.5.8
Fungus resistance	3.4.9	4.6.5.9
Electromagnetic interference	3.4.10	4.6.5.10
Electromagnetic vulnerability	3.4.11	4.6.5.11
Reliability	3.4.12	4.6.6
Maintainability	3.4.13	4.6.7

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:11Z
Check the source to verify that this is the current version before use.



1. PWR connector
2. INPUTS connectors 1 through 6
3. PWR ON/OFF switch
4. Display
5. ELAPSED TIME (HRS) meter
6. PNL LT/DSPL switch
7. DSPL TEST pushbutton
8. INPUT SELECT switch
9. SENSITIVITY SELECT switch
10. MODE SELECT switch
11. FILTER SELECT switch
12. Cover, filter compartment
13. FCTN TEST pushbutton

FIGURE 1. Controls and indicators.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:11Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY
Commanding Officer
Naval Air Engineering Center
Engineering Specifications and Standards Department
(ESSD), Code 93
Lakehurst, NJ 08733

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12503 WASHINGTON, D.C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Commanding Officer
Naval Air Engineering Center
Engineering Specifications and Standards Dept.
(ESSD), Code 93
Lakehurst, NJ 08733

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:11Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-85587	Test Set, Vibration and Cable Case

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording.

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426

PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:11Z
Check the source to verify that this is the current version before use.